

DEEPAK KATUKURI

AI ENGINEER · AGENTIC AI · CLOUD



CONTACT

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- ◆ Potsdam, Germany – open to relocate (Munich)
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TECHNICAL SKILLS

- Languages:** Python, SQL, PL/SQL, Core Java, C++
- LLM & RAG:** Retrieval-Augmented Generation, LangChain, LangGraph, MCP, Langfuse, hybrid retrieval (BM25 + dense, RRF), cross-encoder reranking, pgvector, Chroma, sentence-transformers, embeddings, Ollama
- Cloud & DevOps:** AWS (EC2, RDS, VPC, Elastic IP), Kubernetes (K3s, HPA, Ingress, Secrets), Docker, Docker Compose, GitHub Actions, CI/CD, Vercel, Render
- Deep Learning:** PyTorch, TensorFlow, CNNs, fine-tuning, transformers, Hugging Face, timm, EfficientNet
- Full-Stack:** FastAPI, Flask, React, Vite, Tailwind CSS, PostgreSQL, MySQL, SQLAlchemy, Alembic, JWT, Recharts
- ML & Data:** Scikit-Learn, classification, ensembles, NumPy, Pandas, Matplotlib, Seaborn
- Tools:** Git/GitHub, pytest, Jupyter, Linux (Ubuntu)

EDUCATION

- MSc Data Science**
University of Europe, Germany
Sep 2025 – Present
- BTech in Computer Science & Engineering**
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

- PL/SQL (Database Management)
- Core Java (Software Development)
- OneShield (P&C Insurance)

LANGUAGES

- English – Fluent
- German – A1/A2 (learning)

PROFILE

AI/ML Engineer and Data Science MSc candidate, with 2.5 years of production software engineering behind the ML work. Built a 16-node LangGraph multi-agent pipeline with a Chroma-backed evidence store, a fact-validation gate, and a self-built MCP server. Owned the Kubernetes, CI/CD and AWS layer of a 3-person cloud project, choosing K3s over EKS and shipping autoscaled, zero-downtime rollouts on GitHub Actions. Also built a hand-rolled hybrid RAG retriever with a corrective RAG (CRAG) loop. Looking for English-speaking working student (Werkstudent) AI Engineer roles in Germany – open to relocating to Munich.

PROJECTS

JobPilot: Multi-Agent Job Search & CV Tailoring Pipeline

- Python · LangGraph · Chroma · Typst · github.com/Deepak589/Job_hunt_Agentic_AI
- Built a 16-node LangGraph pipeline that diagnoses a job description, rewrites CV bullets, and blocks any claim without evidence in a Chroma-backed profile store — checkpointed for crash recovery, cost-tracked per run, and held honest by 238 pytest tests across 36 files.

CloudNotes: Kubernetes & AWS Deployment Platform

- AWS EC2 + RDS · K3s · GitHub Actions · Traefik · MSc cloud computing, team of 3 · github.com/Deepak589/cloudnotes
- Replaced a team's manual deploys with a GitHub Actions pipeline to AWS Kubernetes (K3s) — build, push, SSH, zero-downtime rollout — autoscaling 2 to 5 pods at 60% CPU; owned the Kubernetes, CI/CD and AWS layers on a 3-person team.

RAG Pipeline: Hand-Rolled Retrieval to Corrective RAG (CRAG)

- Python · LangChain · pgvector · BGE-M3 · Corrective RAG · github.com/Deepak589/RAG_pipeline
- Built retrieval from scratch through 6 gated stages — hybrid BM25+dense, then framework-validated on LangChain (caught a silent 0.06-recall regression from a default tokenizer) — and added a corrective RAG loop that re-retrieves on a bad match and refuses when the corpus can't answer, reaching 100% correct refusal and 69.5% end-to-end correctness.

RA Nutrition & Lifestyle App: Shipped Full-Stack Product

- React · FastAPI · PostgreSQL · Docker · live at ra-project-indol.vercel.app · github.com/Deepak589/RA_project
- Shipped a full-stack health app — React/Vite frontend on Vercel, FastAPI + PostgreSQL backend on Render — that turns daily meal and symptom logs into a rule-based recommendation and a weekly analytics dashboard, backed by a 99-test pytest suite.

EXPERIENCE

Software Engineer

Mar 2022 – Sep 2024

ValueMomentum · clients: Hiscox, Philadelphia Insurance

- Migrated full policy, claims, premium and customer history for Philadelphia Insurance's Surety Bonds line from a legacy policy admin system into OneShield PAS. Wrote and validated the PL/SQL migration scripts across customer, policy (PID), claims and premium tables.
- Root-caused a recurring data-integrity defect where claims data for pre-2019 Surety bonds under one bond type loaded empty. The OneShield object copy from the stored object to the loading object failed silently when another bond shared that bond type under a different bond ID. Kept records current with a weekly backfill script while a permanent fix was pending.
- Resolved 100+ production defects across two carrier platforms, Hiscox (P&C insurance) and Philadelphia Insurance (surety bonds). Turned underwriting and claims requirements into technical specs, and ran QA/UAT test cases and CI/CD deployments across the full policy lifecycle.

ML Intern

Jun – Jul 2019

SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it on IBM Watson ML. Presented the results to non-technical reviewers with Matplotlib and Seaborn charts.